



Operating system and system programming (25CS2104E)

FILE PERMISSION AUDITOR

Review-I

Under the Esteemed Guidance of

Raghupathi M
CSE Department

Team Members:

Name- P Neeharika Roll Number-2520090233

Name - G Shivani Roll Number-2520090046

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Abstract

Security-focused application for analyzing file and directory access permissions.

Scans a user-selected directory and checks permissions for:

- Owner

- Group

- Other users

Evaluates Read, Write, and Execute permissions.

Identifies unsafe or risky permissions, such as unauthorized write or execute access.

Displays detected permission issues in a clear and understandable format.

Provides an option to correct unsafe permissions for selected files.

Helps prevent accidental modification and unauthorized access to important data.

Demonstrates OS concepts such as:

- File systems

- File permissions

- Ownership

- Access control

- System calls

Provides practical understanding of how operating systems protect files and control access.

Introduction

- File permissions in Linux control access to files and directories and help prevent unauthorized modification or access.
- Incorrect permissions can provide more access than required, creating potential security risks.
- The application performs recursive directory scanning, examining files and subdirectories within the selected directory.
- Based on the access given, files are classified as Safe, Warning, or Critical for easier identification of risks.
- An Audit Log records changes such as the file name, old permission, new permission, and time of change.
- A Permission Report provides a summary of total files scanned, safe files, warnings, and critical permissions.
- The project demonstrates OS concepts including file systems, directory traversal, ownership, access control, system calls, and file handling.

Hardware and Software requirements

List the Requirements in table format

Requirement

Specification

Hardware Requirements

Processor

Intel Core i3 or equivalent

RAM

4 GB

Storage

10 GB free space

Software Requirements

Operating System

Ubuntu/Linux

Programming Language

C

Compiler

GCC

Text Editor

Nano / VS Code

System Architecture

USERS

SELECT
DIRECTORY

RECURSIVE
SCAN

PERMISSION
CHECK

RISK ANALYSIS

SAFE

WARINING

CRITICAL

FILTER RESULTS

FIX
PERMISSIONS

LOG
CHANGES

SYSTEM ARCHITECTURE

: **1. User Input**

→ Select the directory to be scanned.

2. Recursive Scanning

→ Scan the directory and its subdirectories for files.

3. Permission Checking

→ Check read, write, and execute permissions for the owner, group, and others.

4. Risk Analysis

→ Classify files as:

Safe

Warning

Critical

5. Audit Logging

→ Record the file name, old permission, new permission, and time of change.

6. Permission Report

→ Show total scanned files and the number of Safe, Warning, and Critical files.

CONCLUSION

The **File Permission Auditor** provides a simple way to check and understand file and directory permissions in a Linux system. It scans the selected directory, checks the permissions of the owner, group, and other users, and identifies potentially unsafe settings. The project helps demonstrate basic operating system concepts such as **file permissions, ownership, access control, file systems, and system calls**. It also allows unsafe permissions to be corrected when required.

Thank you